

A130 – The Fine-Structure Constant: α as Another Notation for ξ

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.1 The Thesis: In Natural Units $\alpha = 1$

[K] The strongest statement about the fine structure constant is the simplest: in natural units

$$\alpha = 1.$$

The famous value $1/137.036$ is *not* a mystery of nature, but the bookkeeping that arises when converting to SI units – just as $c = 3 \cdot 10^8$ is the bookkeeping of the metre-second choice.

This is not an invention of this theory. Setting α to one is a *known* unit convention: one normalises the charge so that $e^2 = 4\pi$ (rationalised Heaviside-Lorentz units with $\varepsilon_0 = \mu_0 = 1$). In these units $\alpha = 1$ by construction. The theory adopts this convention; it does not re-justify it.

But the reason is different from $c, \hbar, k_B$. For $c, \hbar, k_B$, the conversion factors become one because time-mass duality identifies their dimensions (A085). α is dimensionless; this justification does *not* apply here. The convention $\alpha = 1$ rests on the electromagnetic charge normalisation – a separate reason having nothing to do with time-mass duality and therefore handled here separately, not under the natural-units topic.

.2 How $\alpha = 1$ Is Derived

[B] The derivation runs through the charge normalisation and Maxwell duality, not through an axiom. In SI:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}.$$

With $\hbar = c = 1$ and the natural choice $\epsilon_0 = 1$, the charge normalisation $e^2 = 4\pi$ follows from $\alpha = 1$. The Maxwell relation $c^2 = 1/(\epsilon_0\mu_0)$ with $c = 1$ and $\epsilon_0 = 1$ then gives $\mu_0 = 1$. Substituting confirms:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{4\pi}{4\pi \cdot 1 \cdot 1 \cdot 1} = 1.$$

[K] The core is electromagnetic duality $\frac{1}{\epsilon_0 c} = \mu_0 c$, which is perfectly satisfied in natural units ($1 = 1$). $\alpha = 1$ is the expression of this balance between electric (ϵ_0) and magnetic (μ_0) coupling.

Honest assessment. The step itself is standard: the charge normalisation $e^2 = 4\pi$ is the rationalised convention in which $\alpha = 1$ comes out. What is non-standard is the *interpretation*: that the SI value $1/137$ is thereby not a fundamental mystery of the coupling but measures the historical charge unit. Not everyone shares this interpretation – the classical Duff position holds α to be absolute on account of its dimensionlessness. The theory's actual contribution lies not here, however, but in the next section: that the SI value not only measures charge units but, via $\alpha = \xi E_0^2$, carries the geometric parameter ξ .

This removes the mystification. The question "why exactly $1/137$?" is a question for the SI system, not for nature – of the same kind as "why exactly 299,792,458 metres per second?". The numerical value measures the historical charge unit, not a property of the coupling itself.

.3 What Remains: ξ and One Scale

[K] When all constants are one in natural units, the only genuine content remaining is the single dimensionless parameter ξ and one scale. The SI value of α carries precisely these two quantities:

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2.$$

Here α is no longer the native one but its SI dress. In this dress one sees that α and ξ encode the same quantity, connected by the scale E_0 : ξ is the geometric reading (packing ratio, A020), α_{SI} the electromagnetic one. Two sides of one coin, one degree of freedom – just as mass, energy, temperature, and inverse time are one quantity in four dresses (A085).

.4 Why the Measured Value Is Irrelevant

[K] The theory works with ratios (A080). Ratios are unit-free and depend only on the geometry. The absolute value of $\alpha = 1/137.036$ – is an absolute value and therefore belongs to the bookkeeping layer, not the core.

[K] The *structural relations* at issue are already known and confirmed in SI units and in the Standard Model. The theory does not invent them, it orders them: $\alpha = \xi E_0^2$ connects known quantities. What it asserts is the *structure* of this connection – that α and ξ are the same degree of freedom – not a new numerical value for α .

[B] It follows how the numbers in the following sections are to be read: *not* as a test the theory passes or fails. They show the precision with which the bookkeeping reproduces the SI value – a statement about the bridge constant, not about the structure. As long as one works with ratios, they are irrelevant.

[STIPULATION] The test of the theory lies elsewhere. α is not one of the four prediction quantities of the series (A010); those are ratios and structural statements (m_{τ} , the ξ -order of the CHSH deviation, D_f , L_0). Whether α approaches $1/137$ to three or five digits does not test the theory – it tests the SI conversion.

.5 Why the Circularity Is Not a Weakness

[STIPULATION] As soon as one works in SI dress, the system is circular, and rightly so. Every system that fixes its own constants is circular – the only question is where one begins. (In natural units the question does not arise at all: there $\alpha = 1$ and nothing to determine.)

One can start from ξ (geometric perspective): then ξ is primary and $\alpha = \xi E_0^2$ derived. One can start from α (electromagnetic perspective): then α is primary and $\xi = \alpha/E_0^2$ derived. Both routes give the same values for all other constants. They are mathematically equivalent.

[B] The question “can α be derived from ξ without circularity?” is therefore wrongly posed. In a closed system there is no exterior from which one could derive. There is an entry point one declares, and the rest that follows from it. Classical physics conceals this by placing several constants side by side as “fundamental”; this theory makes explicit that only one entry point is needed.

A mistake would still be possible here, namely presenting the entry point as a derivation from nothing. Whoever substitutes $E_0 = \sqrt{\alpha/\xi}$ and then announces “exactly the measured value of α ” confuses reading consistency with proof. The E_0 determination avoids this mistake, because it does *not* run via $\sqrt{\alpha/\xi}$: E_0 follows from the lepton masses and the K_{frak} derived from ξ (next section), and α is thereby hit forward to 0.017 %. The circularity is therefore broken at this point by the fact that K_{frak} does not come from α . *(Corrected on 30 Sept. 2026: previously “the exact E_0 is reached by Route 2 independently of α , and the agreement of the two α -free routes makes the α point over-determined” – this rested on the formula $E_0^2 = 4\sqrt{2} m_\mu/\xi^p$, which has been removed, see the next section.)*

.6 The Scale E_0 as Entry Point

[K] What separates ξ from α is only the scale E_0 . For it there are two closely related routes, neither of which uses α : the same geometric mean of the lepton masses, once without and once with the fractal correction.

[K] Route 1 (without correction). The geometric mean of the two lightest lepton masses,

$$E_0^{(1)} = \sqrt{m_e m_\mu} = 7.348 \text{ MeV},$$

gives when substituted $\alpha = \xi \cdot (7.348)^2 = 1/138.9$, thus -1.35% from the measured value. This is the variant in which K_{frak} cancels: as a ratio calculation it does not carry the factor (Doc. 122).

[K] Route 2 (with fractal correction). With the factor $K_{\text{frak}} = 1 - 100\xi = 74/75$ derived from ξ in A040,

$$E_0^{(2)} = \sqrt{\frac{m_e m_\mu}{K_{\text{frak}}}} = \frac{7.348}{\sqrt{0.98667}} = 7.397 \text{ MeV},$$

it follows that $\alpha^{-1} = K_{\text{frak}}/(\xi m_e m_\mu) = 137.06$, thus $+0.017\%$ from the measured value; with the rounded anchor $E_0 = 7.398 \text{ MeV}$ one gets 137.035. The square root appears because E_0 enters α quadratically: K_{frak} acts on α^{-1} . *(Corrected on 30 Sept. 2026: Route 2 was previously “a construction from ξ and m_μ alone”, $E_0^2 = 4\sqrt{2} m_\mu/\xi^p$, said to hit 7.398 MeV “without m_e and without back-calculation from α ”. This formula has been removed as a second route: p is derived nowhere, the check script set $p = -0.2679$, which hits exactly 7.398; with $p = 4$ it would give $E_0 = 1.4 \times 10^9 \text{ MeV}$, and it is dimensionally inconsistent (Docs. 011, 041). No derivation exists elsewhere in the corpus either. Route 2 is the fractally corrected mean (corpus R72).)*

.7 The Two Routes Fix K_{frak}

[K] The two routes differ only by K_{frak} . Asking the question the other way round – which factor must stand between them for α to be hit exactly? – the answer is unique:

$$K_{\text{frak}} = \xi m_e m_\mu \alpha^{-1} = \left(\frac{E_0^{(1)}}{\sqrt{\alpha/\xi}} \right)^2 = \left(\frac{7.3479}{7.3980} \right)^2 = 0.98650.$$

[K] This value hits the factor independently derived in A040,

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98667,$$

to $1.7 \cdot 10^{-4}$; read the other way, α requires the coefficient 101.2 instead of 100 in front of ξ . The two routes are thus the indirect proof that K_{frak} can only have this value: A040 obtains it from the cumulative RG run and the consistency condition on the effective fractal dimension, while the α sector requires the same value as the gap between the bare mean and the E_0 that follows from α .

[B] K_{frak} thus has a *second witness* in the α sector. This determination uses α – it is not an α -free measurement –, but it does not use K_{frak} ; what in A040 is a derived constant is here *read off* from α and the lepton masses. (*Corrected on 30 Sept. 2026: previously the two routes were called "mutually independent" and the α point "over-determined"; $K_{\text{frak}} = (7.398/7.348)^{-2} = 0.9862$ was taken as an α -free "measurement" matching $1 - 100\xi$ to $8 \cdot 10^{-5}$, and this residual as a "test of the pole mass". That rested on the removed Route 2 formula. $7.398 = \sqrt{\alpha/\xi}$ contains α ; correct is $(7.3479/7.3980)^2 = 0.98650$, i.e. $1.7 \cdot 10^{-4}$ from $74/75$.)*

.8 A Coincidence That Must Be Resisted

[K] It is tempting to search for a *more elegant* expression for E_0 than the mass route, and there is a striking one: Euler's number squared,

$$e^2 = 7.389 \text{ MeV (?)}, \quad \alpha = \xi \cdot (7.389)^2 = \frac{1}{137.37} \quad (-0.24\%),$$

which hits the SI value even better than the mass route. This expression should be *resisted*, for a principled reason: it equates a pure number (Euler's number) with an energy in the arbitrary unit MeV. One MeV is a human convention; no pure number can fix a physical scale *in* MeV without the agreement depending on the unit choice. In a different energy unit a different number would appear, and the coincidence would vanish.

[K] This is exactly the pattern the corpus warns against elsewhere (A105): a number fitted backwards to an elegant expression without forward content. That e^2 hits closer than $\sqrt{m_e m_\mu}$ does *not* make it the better candidate – the mass route carries a physical reason (the lepton masses), the Euler coincidence carries none. It is recorded here only to explicitly *not* book it as an anchor. The elegant expression is not needed anyway: E_0 is already determined by Route 2 (mass mean with K_{frak}) with forward content, while e^2 merely holds a number to a unit.

.9 Verification Script

- The equivalence of both perspectives ($\alpha = \xi E_0^2$ and $\xi = \alpha/E_0^2$ deliver identical values); the α -free harmonic route with $\kappa = 4$; the two numerical routes; and the explicit check that substituting $\sqrt{\alpha/\xi}$ is a consistency and not an independent proof: `python/A_Serie_Skripte/a130_alpha_kette.py`

.10 Open Questions

First, E_0 is determined α -free, but not over-determined. Route 2 gives $E_0 = 7.397$ MeV from the lepton masses and the K_{frak} derived from ξ , without α ; α then follows to 0.017 %. What remains open is the residual of $1.7 \cdot 10^{-4}$ between the $K_{\text{frak}} = 0.98650$ required by α and 74/75. The Euler coincidence $E_0 = e^2$ remains explicitly *not* a derivation and is not needed. (Corrected on 30 Sept. 2026: previously “The exact $E_0 = 7.398$ is reached by Route 2 from ξ and m_μ , without α and without m_e ... over-determined” – this rested on the removed formula $E_0^2 = 4\sqrt{2} m_\mu / \xi^P$.)

Second, the absolute anchor remains tied to measured masses. Route 2 fixes E_0 from m_e , m_μ and ξ ; the absolute scale therefore enters via the measured lepton masses. A determination of the exact E_0 from purely dimensionless geometry without any mass anchor does not exist – but this is the general structure of the SI bridge (A080), not a deficiency specific to the α sector. Since the theory tests via ratios, the remaining residual carries no weight for its evaluation.

Third, a lower bound applies. The relative description $\alpha \leftrightarrow \xi$ is secured down to the sub-Planck bound $L_0 = \xi \ell_P$ (A210); below this the theory makes no statement.

.11 Supersedes

This document subsumes: Doc. 011 (fine structure constant), Doc. 043 (resolution of constants), Doc. 044, Doc. 101 (circularity of constants, core thesis: one degree of freedom, ξ and α equivalent). Incorporated: P40.

.12 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.